

Steady Engineering Executive Summary

Executive summary

The Steady engineering program should be treated as the construction of **one longitudinal behavioral-health system with three product surfaces**, not as the progressive expansion of a patient chatbot.

Steady Personal is the continuous patient relationship: text and voice interaction, structured memory, validated measurement, interventions, deterministic safety, and per-person learning. **Steady Clinical** converts that same longitudinal history into an explainable, prioritized workflow for clinicians and care managers. **Steady Intelligence** extends the same underlying graph across covered populations, EHR/claims/ADT data, navigation, provider networks, outcomes, utilization, cost, and governed next-action orchestration.

The permanent architecture therefore needs to preserve the chain:

Person → State → Signal → Assessment → Need → Risk → Intervention → Response → Clinical Action → Referral → Encounter → Outcome → Utilization → Cost

The central structural rule is:

Design from the final economic-intelligence state backward. Implement from the foundation forward.

A local implementation decision in an early handoff should not be accepted if it prevents a later requirement without a documented Architecture Decision Record describing the tradeoff, migration path, and affected downstream contracts.

The roadmap is normalized into **nine phases** and five engineering handoffs. Handoff A covers the permanent foundation and Steady Personal. B adds individual learning and Steady Clinical. C adds enterprise interoperability and population intelligence. D adds navigation and governed population learning. E connects outcomes, provider/network performance, utilization, cost, and orchestration.

Competitive research reinforces this structure. Limbic has already demonstrated that a clinical AI companion can combine conversational support, guided CBT and clinician workflows; its official materials report more than 650,000 assessments across 45% of NHS Talking Therapies, while Limbic Care reported supporting more than 55,000 patients between sessions by June 2026. ¹ A 540-person randomized trial found that the AI-enabled CBT application increased engagement relative to digital workbooks, but short-term safety and symptom improvement were similar between groups. ²

That means **conversation alone is not Steady's moat**.

NeuroFlow establishes the other benchmark. Its current enterprise platform identifies behavioral-health risk using claims, EHR, ADT and assessment data; stratifies populations; engages and routes patients; tracks outcomes; and supports referral management and care coordination. ³ Its acquisitions of Owl and Quartet expanded measurement-based care, provider-network and referral/navigation capabilities. ⁴

The resulting Steady thesis is more specific:

Match the patient-continuity capability of leading clinical AI companions. Match the enterprise infrastructure of NeuroFlow-class platforms. Attempt to outperform both by creating a governed longitudinal learning system that combines what happens inside healthcare with what happens between healthcare events.

That outcome is a hypothesis to prove, not a superiority claim to make now.

Product architecture and Phase Zero requirements

The permanent system should use one domain architecture beneath Personal, Clinical and Intelligence. Current-state tables and caches can exist for performance, but they should be projections. The historical meaning of the system should remain reconstructable from durable, versioned events.

Canonical domain entities

Domain	Canonical entities that should exist in the permanent model
Identity	person; patient; user/account; clinician; care manager; provider; organization; payer; employer; care team; tenant; facility; program; contract population; enrollment; external identifier
State and measurement	daily state; clinical state; signal; symptom; mood; anxiety; stress; sleep; functioning; engagement; assessment; instrument; instrument version; response item; score; subscore; severity; trend
Care	need; goal; care plan; intervention; intervention version; protocol; assignment; exposure/completion; response; clinical action; care episode; clinician constraint
AI and evidence	memory; observation; pattern; hypothesis; recommendation; inference; evidence; confidence; model; model version; prompt version; evaluation result; reason code
Safety	rule; rule version; safety state; gate; escalation; crisis event; cooldown; re-entry authorization; safety-policy version
Enterprise and rights	eligibility; benefit; role; permission; policy configuration; organization scope; consent; purpose; data-use right
Healthcare	encounter; referral; appointment; provider-network participation; diagnosis context; medication context; scheduling state

Domain	Canonical entities that should exist in the permanent model
Outcomes and economics	outcome; outcome definition/version; utilization event; claim; billed amount; allowed amount; paid amount; member liability; attribution; episode cost; PMPM period

The most important decision is that **the generated sentence is not the durable clinical/intelligence asset**. The structured evidence behind the sentence is.

Steady therefore needs a permanent truth model that distinguishes:

FACT — explicitly reported by an authorized source.

OBSERVATION — directly measured or recorded.

PATTERN — a derived relationship across observations that can later be contradicted.

HYPOTHESIS — a possible relationship for which evidence is insufficient.

RECOMMENDATION — a candidate action.

DECISION — what the patient, clinician, deterministic system or other authorized actor actually decided.

Those categories should never silently collapse into one another.

Required event registry

The event registry should also be established at the foundation stage, including event families that will not begin emitting production data until later handoffs.

Family	Core events
Identity and rights	person.created, account.linked, enrollment.created, external_id.mapped, consent.granted, consent.changed, consent.withdrawn, data_right.requested, data_right.executed
State and measurement	daily_checkin.completed, state.recorded, assessment.assigned, assessment.completed, assessment.scored, trend.updated
Intervention	intervention.assigned, intervention.started, intervention.completed, intervention.abandoned, intervention.response_recorded, intervention.version_retired
Memory and AI	memory.created, memory.patient_corrected, memory.superseded, ai.inference_requested, ai.inference_completed, ai.output_rejected, pattern.proposed, pattern.contradicted, recommendation.created

Family	Core events
Safety and decisions	safety_rule.triggered , safety_state.transitioned , escalation.created , reentry.authorized , clinician.reviewed , clinician.feedback_recorded , decision.recorded , care_plan.changed
Navigation and healthcare	referral.created , referral.matched , referral.scheduled , referral.failed , referral.closed_loop , encounter.received , appointment.received , eligibility.received
Enterprise and economics	claim.received , utilization.recorded , outcome.measured , metric.calculated , population_score.created , population_score.superseded
Model lifecycle	dataset.manifested , model.evaluated , model.approved , model.deployed , model.flagged , model.rolled_back , drift.detected

Every event should carry canonical subject identity, tenant/organization scope, actor, source, occurred_at , recorded_at , schema version, idempotency key, correlation/causation identifiers, payload version, provenance, and applicable data-use purpose.

For interoperability, Steady should reserve adapter mappings to FHIR and U.S. Core rather than shaping its internal domain around a particular EHR. FHIR is the HL7 standard for exchanging healthcare information electronically, while USCDI and U.S. Core provide standardized U.S. interoperability baselines. ⁵

The internal graph should remain **Steady's domain model**. FHIR, claims, ADT, provider directories and other external representations should enter and leave through adapters.

Nine-phase roadmap

The five coding handoffs map to nine implementation phases:

Phase	Handoff	Purpose	Exit gate
Foundation	A	Permanent event/ data/AI/safety architecture	Six-month simulated journey survives replay, memory correction, consent change, safety transitions and model replacement without schema redesign
Steady Personal	A	Production companion	Continuous text/voice experience operates with typed memory, measurement, intervention-response capture and deterministic safety

Phase	Handoff	Purpose	Exit gate
Individual learning	B	Personal behavioral model	Returning-user personalization can be explained from longitudinal evidence and is measurably different from cold-start behavior
Steady Clinical	B	Supervised clinical workflow	Clinician can identify who needs attention, why, what changed, what has been tried and what happened afterward
Enterprise and interoperability	C	Connect healthcare systems and enterprise identity	Secure, idempotent multi-tenant ingestion works with stable identity reconciliation and source provenance
Population identification and stratification	C	Manage populations larger than active Steady users	Explainable population dimensions operate without collapsing safety, severity, utilization and care-gap risk into one opaque score
Navigation	D	Close the loop to actual care	Need → provider → referral → appointment/failure → outcome can be followed end to end
Population learning	D	Governed multi-patient intelligence	At least one model/ranking task beats a predefined baseline on held-out data and can be reconstructed point-in-time
Outcomes, cost and orchestration	E	Connect care to clinical/economic results	Complete journey reconstructs from original evidence through intervention, human decision, outcome, utilization and cost

Each phase needs both an **engineering gate** and a **measurement gate**.

For the foundation, success means event replay parity, complete provenance, zero duplicate events under retry testing, complete safety-regression performance against the predefined test suite, and verified tenancy isolation.

For Personal, the principal product metrics are activation, sustained retention, intervention completion, patient-rated usefulness, memory correction behavior, model/fallback reliability and structured-output validity. Those measures establish product quality; they are not proof of clinical efficacy.

Individual learning should be judged by human-reviewed pattern quality, intervention-response completeness, contradiction rates and whether longitudinal personalization improves usefulness or intervention completion compared with an appropriate baseline.

Clinical should be measured through clinician-rated usefulness, alert precision, review time, feedback capture, documentation burden and structured evaluation of safety false positives and false negatives.

Enterprise/interoperability should be measured through tenant-isolation testing, ingest/reconciliation success, external-ID accuracy, data freshness and complete transaction traceability.

Population scoring requires calibration, subgroup evaluation, missing-data transparency and stable metric definitions. Navigation requires referral closure, time to care, match quality and explicit failure-reason capture. Population learning adds baseline lift, drift, subgroup performance, rollback readiness and Learning Ledger completeness.

The final economic phase should be judged on outcome completeness, claims linkage, reproducibility of episode/PMPM measures, orchestration acceptance/override behavior and appropriately reviewed economic analyses. A savings claim should not exist merely because a retrospective dashboard displays a cost difference.

Coding handoffs and downstream contracts

The critical engineering idea is that each handoff produces both **features** and **future rights**.

Deliverables comparison

Handoff	Primary deliverables	Durable data/intelligence asset created
A	Canonical graph; event system; tenancy; consent; provenance; AI Gateway; memory; intervention ontology; safety; adapter contracts; Steady Personal	Longitudinal evidence history
B	Personal learning; intervention-response engine; Clinical; prioritization; feedback/evaluation	Patterns, hypotheses, recommendation history and clinician labels
C	Enterprise identity; SSO/SCIM; FHIR/ADT/claims; population scoring; measurement-based care	External clinical/utilization history plus historical population scores
D	Provider graph; closed-loop referral; navigation; model registry; governed analytical datasets; Learning Ledger	Provider/referral outcomes and accountable population-learning history
E	Outcomes; utilization; cost; provider/network intelligence; pathways; orchestration	Full evidence-to-outcome-to-cost graph

Acceptance-test comparison

Handoff	Required acceptance evidence
A	Six-month longitudinal replay; idempotent retries; memory correction; consent change/ withdrawal; safety state/re-entry; model-vendor replacement; permission/provenance tests; adapter contract tests

Handoff	Required acceptance evidence
B	Patterns reconstruct from evidence; contradiction works; abandonment/non-response retained; recommendation differs from selected action; prioritization is explainable; clinician feedback is queryable
C	Cross-tenant attack suite passes; ingestion is idempotent; raw source is preserved; instruments/scores replay by version; population score history remains queryable
D	Referral closes end-to-end; deterministic constraints cannot be bypassed by AI; model prediction reconstructs point-in-time; no future-data leakage; rollback succeeds
E	Evidence-to-cost journey reconstructs; attribution/outcome versions replay; PMPM calculations reproduce; provider scores expose components; safety overrides economic optimization

Downstream-contract comparison

Handoff	What must be preserved for later
A	Stable identity, tenant scope, event semantics, provenance, external IDs, integration interfaces, model independence, intervention versions, data rights, future outcome/cost linkage
B	Explainable scoring components, clinician feedback labels, action/outcome history, external identifiers on care objects
C	Time-bounded provider/network structures, raw and normalized external data, population score history, claims/utilization lineage
D	Referral/provider outcomes, dataset/model versions, point-in-time predictions, causal caveats, cost/outcome linkage
E	Should consume the semantics already created; E should not force a new foundational data model

The primary architectural failure to avoid is reaching E and discovering that A stored only chat transcripts and mutable "current status" fields.

Competitive position, learning architecture and AI governance

Limbic confirms that a clinically constrained AI relationship layer is becoming a real care-delivery capability. Limbic Care offers on-demand conversational support and guided CBT, while Limbic Access supports intake, clinical decision support, risk workflows and EHR interoperability. ⁶ Limbic also describes a proprietary clinical layer between users and general-purpose LLMs that evaluates inputs, constrains interactions and screens outputs. ⁷

Therefore, Steady should **not** claim that putting deterministic or clinical safeguards around an LLM is, by itself, unique.

Steady's stronger target differentiation is the structured longitudinal model:

- What changed for this person?
- What preceded it?
- What did Steady or the care team do?
- What did the person actually do?
- What happened immediately afterward?
- What happened later?
- Did the pattern repeat?
- Was Steady's inference ultimately correct?

NeuroFlow sets the broader enterprise requirement. Its platform currently combines population identification, acuity-based routing, digital engagement, longitudinal outcome measurement and referral management using multiple healthcare data sources. ⁸ Its Owl acquisition added deep measurement-based-care capability, while Quartet added behavioral-health provider-network and referral-management assets. ⁴

A peer-reviewed 2025 American Journal of Managed Care analysis of the Quartet/Independence Blue Cross intervention reported a reduction in total medical PMPM and lower behavioral-health inpatient and emergency utilization relative to a risk-adjusted comparison cohort. The paper also discloses relevant employment and company relationships, so the appropriate conclusion is that risk-bearing purchasers value measurable care access, utilization and economics—not that Steady can assume the same savings. ⁹

The two learning systems

Per-person learning remains tied to one person's own history. It should learn and update relationships among state, timing, context, interventions, preferences and measured response.

Population learning operates across governed cohorts. Its purpose is to determine whether longitudinal information improves defined prediction, ranking or routing tasks across people.

Those systems cannot be allowed to blur privacy and scientific boundaries. HHS guidance confirms that HIPAA obligations continue when regulated entities and business associates use cloud services for ePHI.

¹⁰ Operational data access should therefore not be treated as automatic authorization for model-development use.

The Learning Ledger

Every important model inference should answer eight questions:

Question	What must be retained
What did Steady think?	Prediction, pattern, ranking or recommendation; model/version; prompt/policy version

Question	What must be retained
Why?	Evidence IDs, feature snapshot/reason codes, retrieval set, data freshness, confidence/calibration
What did it show?	Output reference/hash, intended audience, authority tier and warnings
What action followed?	Patient choice, clinician decision, deterministic action, referral or no action
What happened afterward?	Defined outcome window and subsequent events
Was the inference right?	Evaluation result and error taxonomy
Did a human agree?	Review, override, escalation or care-plan change
Can it be reproduced?	Exact model, prompt, policy, dataset and point-in-time evidence state

That ledger is both an engineering asset and part of the eventual moat. It lets Steady learn not only from what happened but from **whether Steady's own prior intelligence was useful and accurate**.

NIST's AI RMF treats AI risk management as a lifecycle process involving governance, mapping, measurement and management. ¹¹ FDA's current AI-enabled-device guidance likewise emphasizes lifecycle risk management and documentation where the software falls within device scope, and current Good Machine Learning Practice principles emphasize safe, effective and high-quality development throughout the product lifecycle. ¹²

In architecture terms, that means model and prompt registries, dataset manifests, evaluation gates, model flags, deployment approvals, subgroup testing, drift monitoring, rollback, and point-in-time reconstruction should be designed before the first population model exists.

Highest-priority decisions for Handoff A

These are the decisions most likely to cause destructive rework if left unresolved.

Priority	Decision	Requirement now	What fails later if omitted
P0	Database/ event model	Choose authoritative event/write model, event envelope, ordering, event time vs record time, idempotency, schema evolution, replay and projections	Population history, claims reconciliation and point-in-time model evaluation become unreliable
P0	Provenance	One shared provenance envelope across AI, safety, measurement, memory, clinicians, integrations, outcomes and economics	Audit, Learning Ledger and economic attribution fragment

Priority	Decision	Requirement now	What fails later if omitted
P0	AI Gateway	One model-call contract with task IDs, structured outputs, tool permissions, PHI purpose, model/prompt version, evidence, fallback and evaluation	Every feature becomes coupled to specific vendors and prompts
P0	Memory schema	Typed, versioned memory with sensitivity, provenance, scope, patient control and model-use permissions	Chat history becomes memory; correction and data-rights handling become difficult
P0	Intervention ontology	Separate Intervention from InterventionVersion; capture target, contraindications, exposure and immediate/delayed response	Personal/population intervention learning becomes scientifically weak
P0	Adapter contracts	Define EHR/FHIR, ADT, claims, provider directory, eligibility, referral, scheduling, outcome and cost interfaces now	C-E require domain rewrites instead of adapter implementations
P1	Identity and tenancy	Separate person, account, enrollment, organization membership and external IDs	Consumer-to-enterprise migration produces duplicate identity and attribution problems
P1	Consent and data purpose	Separate operational, analytics, research and model-development purposes	Later learning programs lack defensible data-use boundaries
P1	Safety state machine	Explicit deterministic states, rules, versions and re-entry logic	Model output can accidentally acquire safety authority
P1	Outcome/cost linkage	Reserve stable episode, provider, outcome, utilization and cost links now	E requires destructive historical migrations
P1	Observability	Correlate user action → command → event → AI → safety → integration → clinician workflow	Incidents and model recommendations cannot be reconstructed
P1	Model/dataset lifecycle	Establish version registries and deployment identifiers now	D adds a separate governance stack incompatible with early history

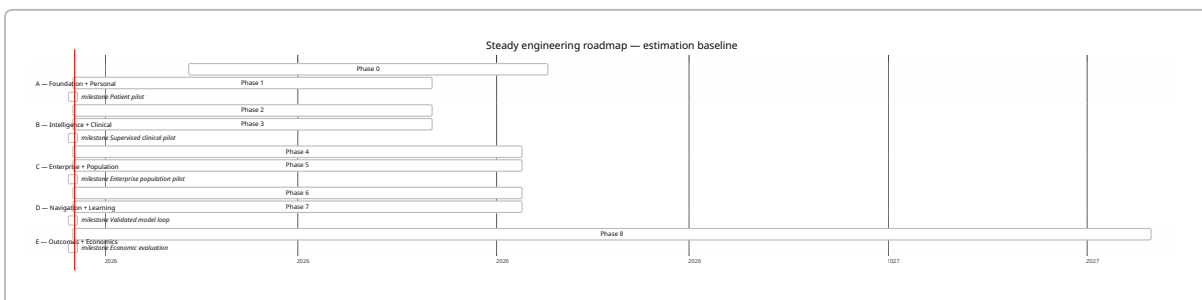
The recommended implementation pattern remains a **modular monolith initially**, rather than premature microservices. The essential requirement is not distributed deployment; it is stable domain boundaries. Identity, state, measurement, intervention, memory, safety, AI, consent, integration and audit should be separable modules with explicit interfaces so later extraction does not alter domain meaning.

Timeline, packaging and engineering files

The timeline is a planning baseline for dependency and estimation discussions, **not a delivery commitment**. It assumes a senior product engineering team with dedicated AI/ML capacity and access to clinical, security/privacy, data and interoperability expertise.

Workstream	Months 0–4	Months 4–8	Months 8–13	Months 13–18	Months 18–24
A — Foundation + Personal	Foundation + companion				
B — Personal Intelligence + Clinical		Learning + supervised clinical pilot			
C — Enterprise + Population			Interoperability + population pilot		
D — Navigation + Population Learning				Closed-loop referral + validated model	
E — Outcomes + Cost + Orchestration					Economic intelligence
Major milestone	Patient pilot	Supervised clinical pilot	Enterprise population pilot	Closed-loop care + model validation	Outcome/cost evaluation

The corresponding Mermaid source is included in the PDF:



I produced two engineering-ready files. The standalone executive summary is **13 pages**, including the one-page timeline, A–E comparison tables, Phase Zero entity/event definitions, competitor analysis, Learning Ledger, governance requirements and the prioritized Handoff A checklist.

[Download the standalone Steady Engineering Executive Summary](#)

The second file prepends that summary to the existing **00 + A-E Master Engineering Handoff Series**, creating a single **27-page engineering packet** in the intended reading order.

[Download the complete Steady Engineering Handoff Pack](#)

The recommended file pairing is therefore:

`Steady_Engineering_Executive_Summary_2026-08-27.pdf` — architecture review, dependency analysis and work-estimation entry point.

`Steady_Engineering_Handoff_Pack_2026-08-27.pdf` — Executive Summary → Master Architecture → A → B → C → D → E.

The practical gate before coding begins is straightforward: **A should not be approved solely by the developers building A**. The data/platform, AI/ML, clinical-safety, security/privacy and interoperability owners should all review A specifically for whether it preserves the contracts required by C, D and E. That same downstream-contract review should occur at every subsequent handoff boundary.

1 <https://www.limbic.ai/nhs-talking-therapies>

<https://www.limbic.ai/nhs-talking-therapies>

2 <https://pubmed.ncbi.nlm.nih.gov/41540194/>

<https://pubmed.ncbi.nlm.nih.gov/41540194/>

3 <https://www.neuroflow.com/health-systems/>

<https://www.neuroflow.com/health-systems/>

4 <https://www.neuroflow.com/neuroflow-acquires-owl-creating-the-largest-end-to-end-platform-for-behavioral-health-measurement-and-engagement-across-the-care-continuum/?noamp=mobile>

<https://www.neuroflow.com/neuroflow-acquires-owl-creating-the-largest-end-to-end-platform-for-behavioral-health-measurement-and-engagement-across-the-care-continuum/?noamp=mobile>

5 <https://isp.healthit.gov/united-states-core-data-interoperability-uscdi>

<https://isp.healthit.gov/united-states-core-data-interoperability-uscdi>

6 <https://www.limbic.ai/care>

<https://www.limbic.ai/care>

7 <https://www.limbic.ai/layer>

<https://www.limbic.ai/layer>

8 https://www.neuroflow.com/?28ee73ed_page=4&3da07ef0_page=3

https://www.neuroflow.com/?28ee73ed_page=4&3da07ef0_page=3

9 <https://www.ajmc.com/view/linking-insured-adults-to-behavioral-health-care-a-cost-saving-solution>

<https://www.ajmc.com/view/linking-insured-adults-to-behavioral-health-care-a-cost-saving-solution>

¹⁰ <https://www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html>

<https://www.hhs.gov/hipaa/for-professionals/special-topics/health-information-technology/cloud-computing/index.html>

¹¹ <https://www.nist.gov/itl/ai-risk-management-framework>

<https://www.nist.gov/itl/ai-risk-management-framework>

¹² <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/artificial-intelligence-enabled-device-software-functions-lifecycle-management-and-marketing>

<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/artificial-intelligence-enabled-device-software-functions-lifecycle-management-and-marketing>